

SECURE CHAT APPLICATION



SEGi
University &
Colleges

University of
Hertfordshire **UH**

TAN JIAJIAN
SCSJ1901781

SV: Ms. Nur Diana Madinah



ChipChat is a secure web-based chat application that implements End-to-End Encryption (E2EE) using RSA-OAEP and AES-GCM to protect user communication. The system provides secure real-time messaging, media sharing and user authentication while integrating cybersecurity features such as SQL injection detection, XSS detection, brute-force attack monitoring. Testing results demonstrated secure, reliable and efficient communication for modern digital environments.

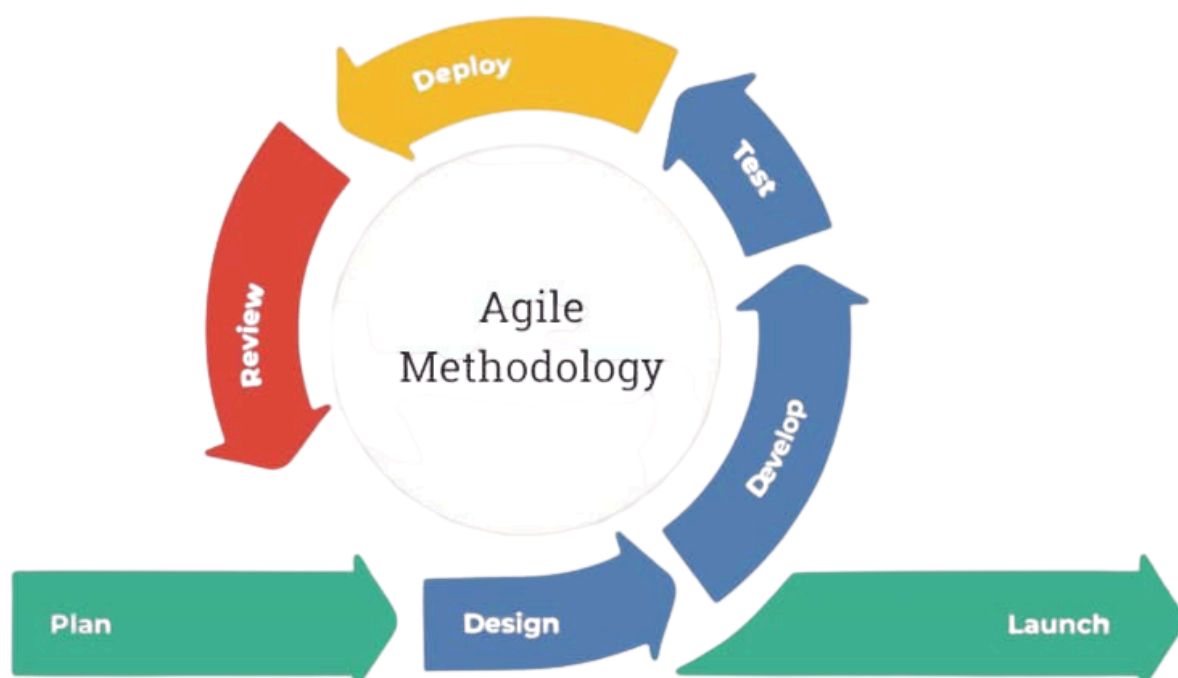
Problem Statement

- Lack of Security in Communication Platforms
- Privacy Concerns and Data Breaches
- Need for Reliable Encryption and Key Management

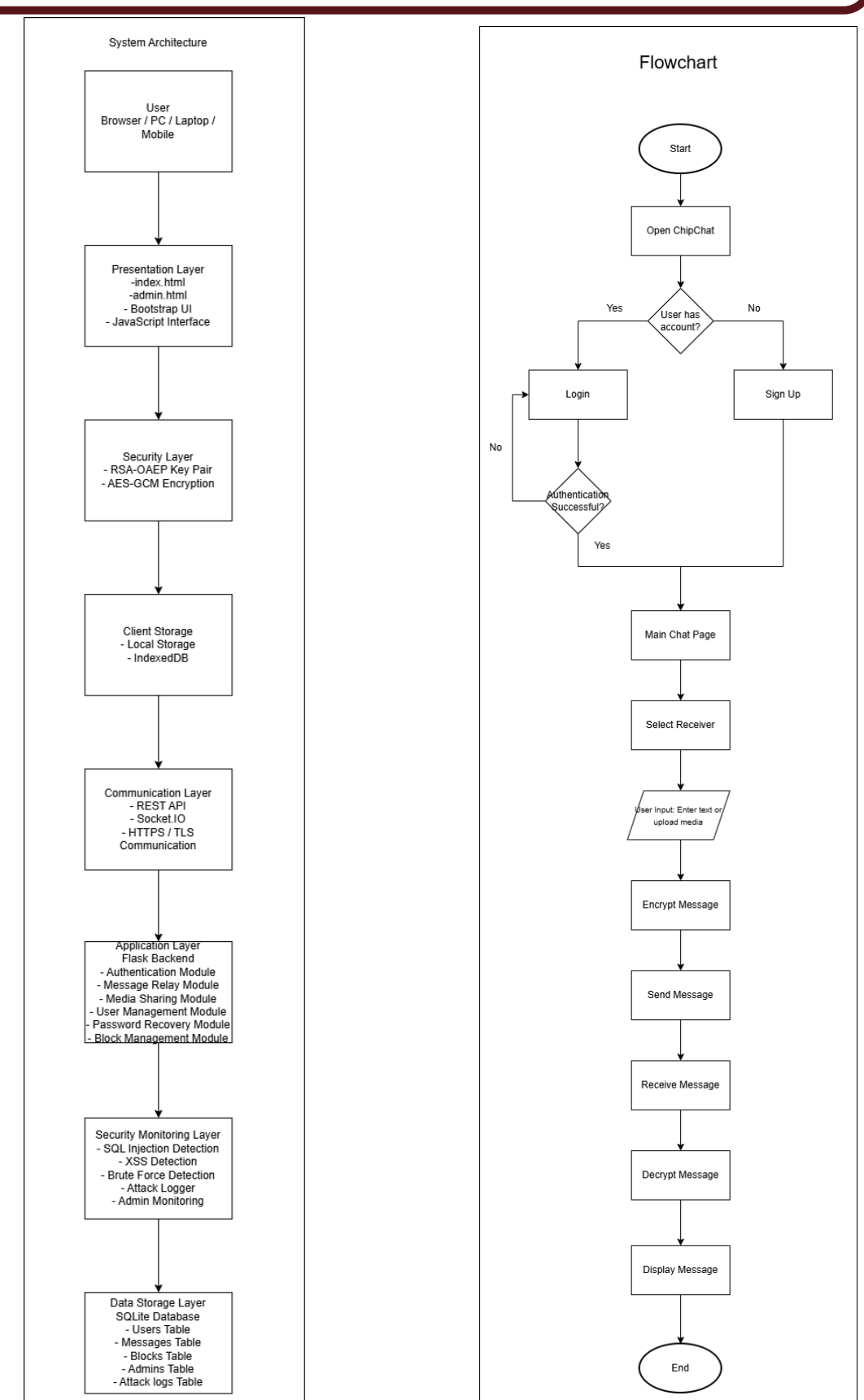
Objectives

- To develop a secure chat application using RSA-OAEP and AES-GCM cryptographic algorithms with end-to-end encryption.
- To ensure message confidentiality, integrity, and authentication.
- To evaluate performance based on the encryption speed, delivery time, and usability.

Methodology



Architecture Flowchart



Project Outcome

- A fully functional web-based secure chat application capable of transmitting encrypted messages and media between users in real time using end-to-end encryption.
- Successful implementation of RSA-OAEP and AES-GCM cryptographic mechanisms to ensure that message contents remain accessible only to authorized communication participants.
- Improved communication security through the integration of user authentication, password hashing, encrypted user identifiers and cybersecurity protection mechanisms such as SQL injection detection, XSS detection and brute-force attack monitoring.
- A practical demonstration of how cryptography, web technologies and cybersecurity principles can be integrated to develop a secure and user-friendly communication platform.

Conclusion

ChipChat successfully demonstrates secure real-time communication through the implementation of end-to-end encryption using RSA-OAEP and AES-GCM. The system protects user privacy, ensures message security and incorporates cybersecurity protection mechanisms such as attack detection and monitoring. The project highlights the practical application of cryptography in modern communication systems.